Cisco PE Node MAR-PE-13 Optical Transport Failure Response Procedure

Node: MAR-PE-13 | **Vendor:** Cisco | **Severity:** Warning | **Alarm Code:** ALARM_2750

Description

This Standard Operating Procedure (SOP) outlines the steps to identify, isolate, and resolve Optical Transport Failures on Node MAR-PE-13 at the Marathahalli Core site, utilizing Cisco vendor-specific equipment and CLI dialect. The failure domain pertains to SFP/XFP degradation, fiber cuts, and Rx/Tx power loss, manifesting as ALARM_2750 with Warning severity. The Optical Transport Network (OTN) utilizes a Wavelength Division Multiplexing (WDM) layer to transport multiple wavelengths of light across the Optical Transport Network. The degradation or failure of an SFP/XFP, or a fiber cut, may cause a loss of signal quality or complete loss of signal, resulting in the ALARM_2750 warning. Furthermore, the SFP/XFP degradation may be due to a faulty or degraded optical interface, while a fiber cut may be caused by human error or physical damage to the fiber cable.

Pre-requisite Checks

- item: Confirm dispatch of remote hands to Marathahalli Core for assistance, note: Ensure all necessary tools and equipment are available and accessible

- item: Verify the status of redundant links and ensure they are operational, note: Check for any existing configuration or software issues

- item: Obtain access to the Cisco device CLI, note: Ensure the necessary credentials and permissions are in place

Potential Root Causes

Cause	Probability
SFP/XFP degradation	High
Fiber cuts	Medium
Rx/Tx power loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface gigabitethernet 0/0/0, Expected Output: GigabitEthernet0/0/0 is up, line protocol is up (connected), Description: This command is used to verify the link status of the GigabitEthernet0/0/0 interface, which may be affected by the Optical Transport Failure. The 'show interface' command provides detailed information about the interface, including its operational status, line protocol status, and any errors that may be present.

Command: show controllers gigabitethernet 0/0/0, Expected Output: GigabitEthernet0/0/0 Controller: 0x6000001, Description: This command
```

```
is used to verify the controller status of the GigabitEthernet0/0/0 interface, which may be affected by the Optical Transport Failure. The 'show controllers' command provides detailed information about the controller, including its operational status and any errors that may be present.

Command: show log | include ALARM_2750, Expected Output: ALARM_2750: Optical transport failure detected on interface GigabitEthernet0/0/0, Description: This command is used to verify the presence of the ALARM_2750 warning in the system log. The 'show log' command provides detailed information about the system log, including any errors or warnings that may be present.
```

Resolution Steps

Step 1: Step: Replace faulty SFP/XFP, Command: no interface gigabitethernet 0/0/0, Description: This step is used to remove the faulty SFP/XFP from the interface. The 'no interface' command is used to disable the interface and remove the SFP/XFP.

Step 2: Step: Replace fiber cable, Command: interface gigabitethernet 0/0/0, Description: This step is used to replace the fiber cable and re-establish the connection. The 'interface' command is used to enable the interface and re-establish the connection.

Post-Check Verification

```
Command: show interface gigabitethernet 0/0/0, Expected Output: GigabitEthernet0/0/0 is up, line protocol is up (connected), Description: This command is used to verify that the link status of the GigabitEthernet0/0/0 interface is operational and connected.

Command: show log | include ALARM_2750, Expected Output: ALARM_2750: No optical transport failure detected on interface GigabitEthernet0/0/0, Description: This command is used to verify that the ALARM_2750 warning is no longer present in the system log.
```

Roll-back Procedures

- Step: Restore original SFP/XFP, Command: interface gigabitethernet 0/0/0, Description: This step is used to restore the original SFP/XFP and re-establish the connection. The 'interface' command is used to enable the interface and re-establish the connection.

- Step: Restore original fiber cable, Command: no interface gigabitethernet 0/0/0, Description: This step is used to restore the original fiber cable and re-establish the connection. The 'no interface' command is used to disable the interface and remove the SFP/XFP.